INTEGRATED SCIENCES
TSM 03
21/11/2018 8.30 AM - 11.30 AM

Rwanda Education Board

TTC NATIONAL EXAMINATIONS, 2018

SUBJECT: INTEGRATED SCIENCES

OPTION: SCIENCES AND MATHEMATICS EDUCATION (TSM)

DURATION: 3 HOURS

INSTRUCTIONS:

1) Write your names and index number on the answer booklet as written on your registration form and **DO NOT write** your names and index number on additional answer sheets of paper if provided.
2) Do not open this question paper until you are told to do so.
3) This paper consists of **two** sections: **A and B.**
   - **Section A**: Attempt **ALL** questions. **(55 marks)**
   - **Section B:** Attempt any **THREE** questions. **(45 marks)**
4) Use only a **blue** or **black** pen.

SECTION A: ATTEMPT ALL QUESTIONS. (55 Marks)

1) Explain the reason why the metallic character of elements increases as we move downwards in a group. **(2 marks)**

2) Write a balanced chemical equation of the reaction between sodium (Na) and water (H_2O). **(1 mark)**

3) Given the following compounds: NaCl, PCl_3, $CaCl_2$ and $SiCl_4$, classify the compounds into those which are covalent and ionic. **(2 marks)**

4) A compound contains 40% of carbon, 6.6% of hydrogen and the rest is oxygen. If the molecular mass of the compound is 180, determine:

   (a) The empirical formula of the compound. **(2 marks)**

   (b) The molecular formula of the compound. **(2 marks)**

5) (a) An isotope G has a half-life of 2.5 hours. What percentage of a given mass of isotope G will be left after 10 hours? **(3 marks)**

   (b) Name two substances that are responsible for acid rain. **(1 mark)**

6) (a) State Boyle's law. **(2 marks)**

   (b) A cylinder of 2 litres by volume contains air at a pressure of 200.000 Nm^{-2} .

   The piston is compressed such that the air in the piston occupies a volume of 0.6 litre.

   Calculate the new pressure of the air in the cylinder if the temperature is kept constant. **(3 marks)**

7) Complete the table below: **(4 marks)**

ANIMAL	PHYLUM	CLASS
Human		
Toad		
Snake		
Cockroach		

8) The flow chart below represents the breakdown of starch in the human gut.

Starch $\xrightarrow{\textit{Amylase}}$ Maltose $\xrightarrow{\textit{Maltase}}$ Glucose

a) Name two organs which produce amylase in humans. **(2 marks)**

b) Describe how the release of amylase from each of these organs is controlled.

(2 marks)

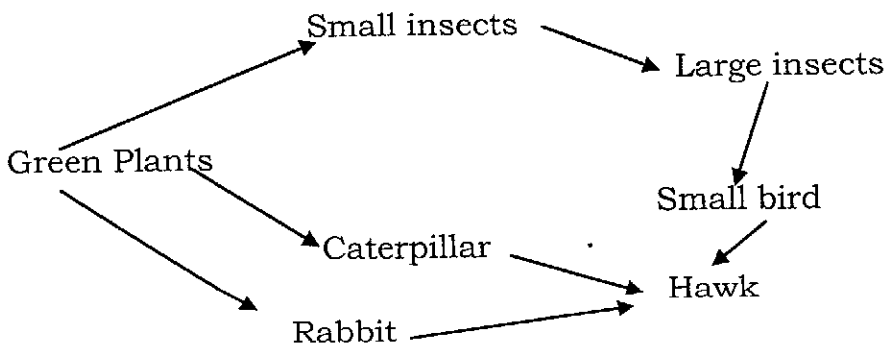
9) Suggest reasons for unintended pregnancy prevention. **(2 marks)**

10) Abstinence is the most efficient method for young people to avoid pregnancy. Explain why. **(2 marks)**

11) Copy and complete the table below by giving four differences between cellulose and glucose. **(4 marks)**

CELLULOSE	GLUCOSE
I.	
II.	
III.	
IV.	

12) The diagram below shows a food web.



Using the information in the food web, construct two food chains. **(2 marks)**

13) Copy and complete the following table: **(3 marks)**

Physical quantity	Name of SI unit	Symbol of SI unit
		N
Displacement		
	Hertz	

14) Explain the term global warming and suggest any two global warming solutions. **(3 marks)**

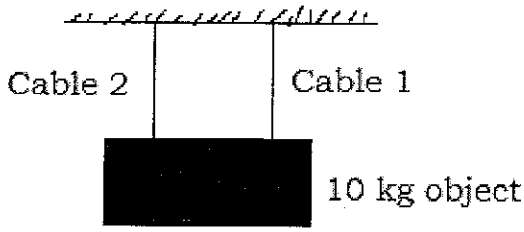
15) Differentiate between light waves and sound waves. **(2 marks)**

16) (a) A thin lens has a focal length f = - 10 cm .Name this lens. **(1 mark)**

(b) Where should an object be placed so that a real and inverted image of the same size is obtained using a convex lens? **(1 mark)**

(c) What happens when a ray of light passes through the optical centre of a diverging lens? **(1 mark)**

17) An object of 10 kg is suspended by two identical cables. It is in static equilibrium. Take g = 9.81 m/s^2.



(a) Determine the weight of the object. **(2 marks)**

(b) Find the tensions T_1 and T_2 in the cables 1 and 2 respectively. Justify your answer. **(2 marks)**

18) Apply Newton's laws of motion to answer "True" or "False" and explain your answer or give an example proving the statement true or false.

(a) If an object is accelerating, a force is acting on it and its speed increases with time. **(2 marks)**

(b) If an object isn't moving, no external force acts on it. **(2 marks)**

SECTION B: ATTEMPT ANY THREE QUESTIONS (45 Marks)

19) In the extraction of iron, the ore is mixed with limestone and coke and then they are heated in a blast furnace.

(a) State the role of:

(i) Coke in the mixture. **(1 mark)**

(ii) Limestone in the mixture. **(1 mark)**

(b) Explain the importance of slag that floats on molten iron. **(2 marks)**

(c) State two characteristics of transition metals. **(2 marks)**

(d) Describe two methods that can be used to prevent rusting of iron. **(2 marks)**

(e) C_5H_{12} is an organic compound which is a member of the homologous series of alkanes.

(i) Give the IUPAC name of C_5H_{12} with a linear chain. **(1 mark)**

(ii) Give the molecular formula of an alkane with 7 carbon atoms. **(1 mark)**

(iii) Alkanes are examples of fossil fuels. Explain one environmental problem caused by the burning of alkanes. **(2 marks)**

(f) An alcohol has the molecular formulae of C_3H_80:

(i) Write the formula of the functional group in alcohols. **(1 mark)**

(ii) Write the structural formulae of two possible isomers of C_3H_80 alcohol. **(2 marks)**

20) Write a short essay on the ecological aspects of each of the following:

(a) Conservation **(5 marks)**

(b) Deforestation **(10 marks)**

21) (a) What is variation in living things? **(2 marks)**

(b) Explain how Meiosis contributes to genetic variation. **(3 marks)**

(c) State five ways in which a human being can differ from another. For each case, indicate whether you think the variation (difference) is due to genes, environment or both. **(10 marks)**

22) (a) The following is the list of some electronic components:

Capacitor, inductor, diode and transistor.

(i) Give the symbols of a capacitor, an inductor, a diode and transistor. **(4 marks)**

(ii) What happen when the diode is reverse-biased in closed electrical circuit? **(1 mark)**

(b) You are supplied with 4 identical dry cells, each having the electromotive force (e.m.f) of 1.5 V and an internal resistance of 0.3 Ω, a voltmeter and a resistor of 2.8 Ω.

(i) Design an electric current circuit comprising 4 dry cells connected in series with the resistor of 2.8 Ω and the voltmeter to measure the potential difference across the given resistor. **(2 marks)**

(ii) Find the overall e.m.f of the circuit. **(2 marks)**

(iii) Calculate:

(1) The total resistance of the circuit. **(2 marks)**

(2) The current flowing through the circuit. **(2 marks)**

(3) The potential difference across the resistor of 2.8 Ω. **(2 marks)**

23) Copy the following table and use the kinetic theory of matter to complete the properties of the three states according to some general properties of matter given below. **(15 marks)**

Properties	**Solid**	**Liquid**	**Gas**
Volume			
Shape			
Spaces between particles (atoms or molecules)			
Particle movement			
Particle organization			

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